Brain Condition Detector SoC

An Event-Driven Heterogeneous System-on-Chip for Multi-Modal Neurological Screening with Cascaded RTL Thresholding, Edge-AI and Swin-Transformer Acceleration

Architecture, AMBA AXI4 IP Integration and Biosensor Specification | November 2026

1. ABSTRACT

Neurological disorders, spanning epilepsy and sleep disorders through neurodegenerative disease, stroke and intracranial lesions, are a leading cause of disability worldwide, yet continuous quantitative monitoring remains confined to hospital laboratories. Existing acquisition systems are bulky, power-hungry and dependent on cloud offload, which limits ambulatory use and raises latency and privacy concerns. This work presents the architecture of the **Brain Condition Detector (BCD)**, a heterogeneous, event-driven System-on-Chip that integrates a multi-channel biopotential analog front end, a high-resolution ADC, a streaming digital signal-processing pipeline and a two-tier inference hierarchy on a single die.

Acquired signals traverse an AFE buffer, a 24-bit ADC, a DSP buffer, a programmable digital filter bank and a ring-organised SRAM buffer before reaching a lightweight RTL thresholding engine. This engine continuously evaluates line-length, band-power-ratio and spectral-entropy criteria at microwatt-class power. Windows judged *normal* are bypassed to a compressed low-rate output. Windows judged *abnormal* assert a wake-up that power-ungates two parallel accelerators: a quantised Edge-AI network for low-latency event classification, and a Swin Transformer engine employing shifted-window self-attention for fine-grained spatio-spectral analysis of brain data. Both accelerators contribute to a diagnostic image (topographic or saliency map) and a textual problem report.

The SoC is partitioned entirely on AMBA AXI4: **AXI4-Stream** carries continuous sample data between pipeline stages, **AXI4-Full** carries burst traffic between DMA engines, memories and accelerators, and **AXI4-Lite** provides the control-and-status plane for every block. By gating high-energy inference behind an always-on screening stage, the architecture is intended to reduce average power substantially relative to always-on neural inference; this is a design objective to be quantified through FPGA prototyping and post-layout power analysis. The system is positioned as a screening and clinical decision-support device, not a stand-alone diagnostic instrument.

Keywords: System-on-Chip, EEG, biopotential AFE, event-driven wake-up, Edge AI, Swin Transformer, hardware accelerator, AMBA AXI4, AXI4-Stream, neurological disorders.

Scope note. Scalp EEG and the other on-body signals in this design can screen for or characterise epilepsy, sleep disorders, encephalopathy, dementia-related slowing, concussion, and stroke-related asymmetry. They cannot, on their own, diagnose brain tumours or structural lesions. For that pathway the Swin accelerator is fed MRI/CT slices through a host interface (Section 5, Tier 3), and the SoC output should be treated as triage support. Any clinical claim needs validation and regulatory clearance (IEC 60601-1 / 60601-2-26, IEC 62304, ISO 14971, and SaMD pathways such as CDSCO or FDA).

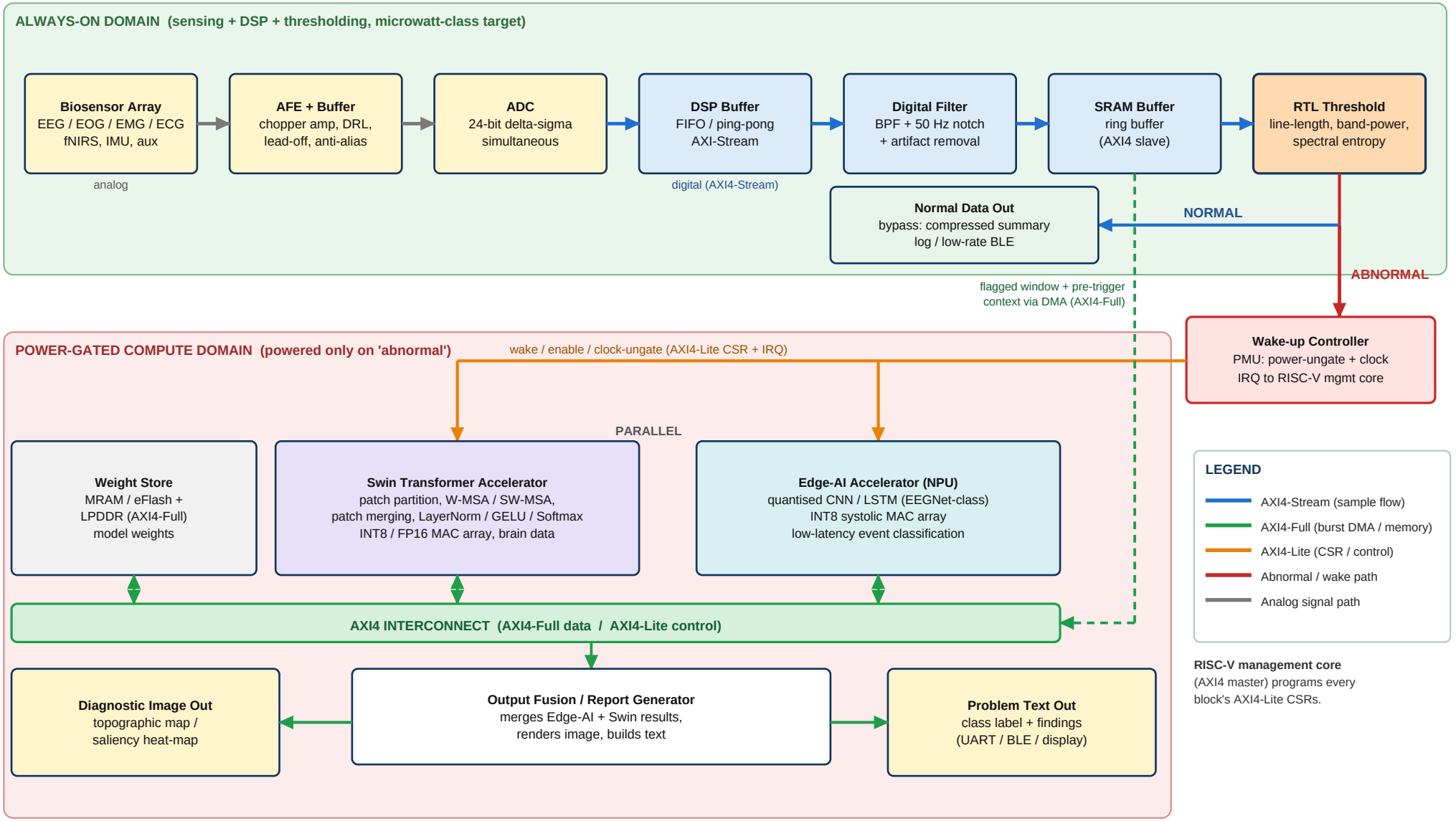
2. SYSTEM ARCHITECTURE AND DATA FLOW

Figure 1 (next page) shows the end-to-end data flow. The chain is split into two power domains. The **always-on domain** contains sensing, conversion, filtering, buffering and the RTL threshold engine; the **power-gated compute domain** contains both accelerators, the output fusion block and their memory traffic, and is powered up only when the threshold engine flags a window. Figure 2 shows how the same blocks are tied together using the three AXI4 protocols.

2.1 Stage-by-stage description

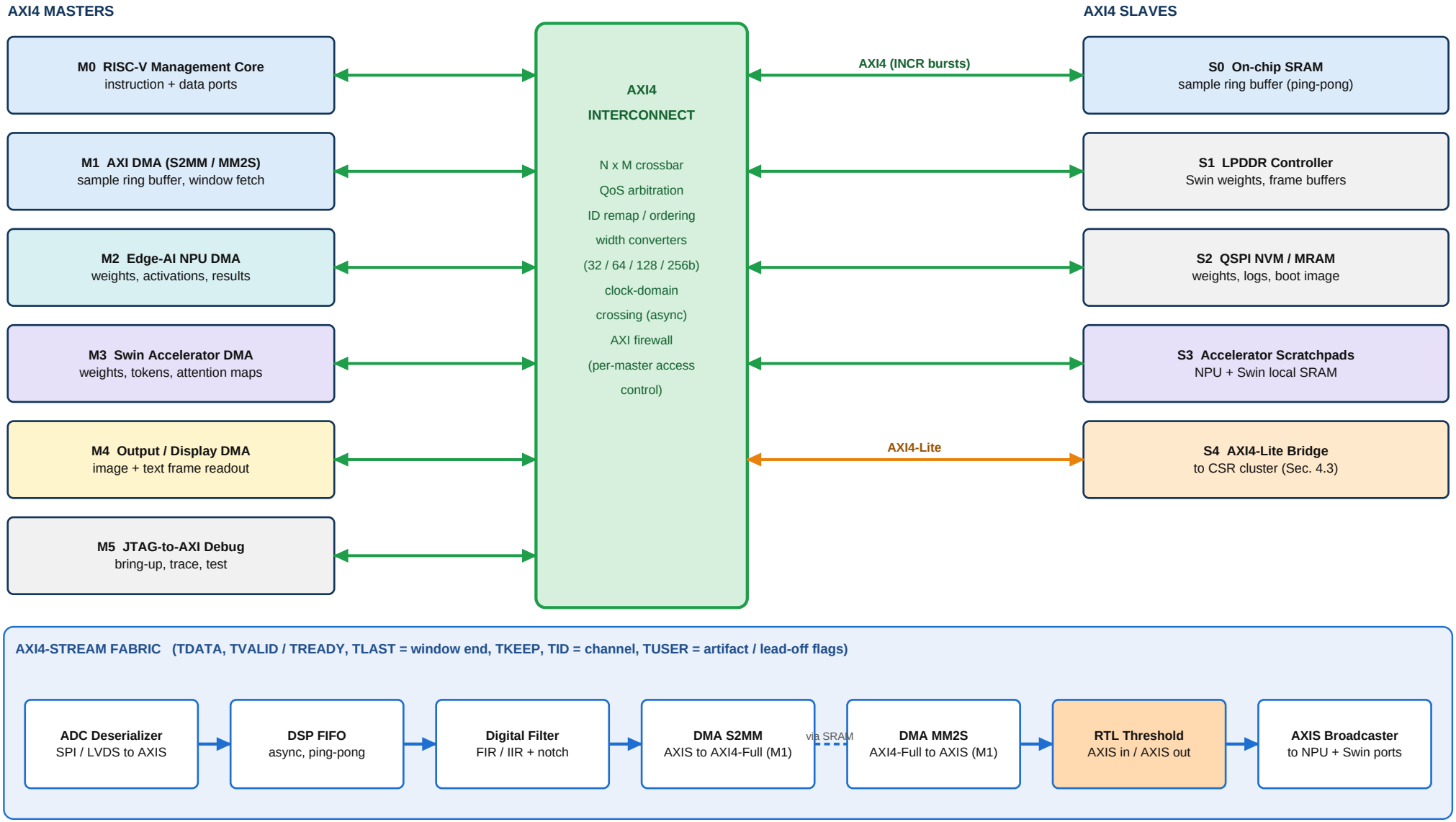
Stage	Function	Key parameters (design targets)
Biosensor array	Captures biopotential, optical, motion and auxiliary signals (Section 5).	Typ. 19 to 32 EEG channels plus EOG / EMG / ECG and auxiliary channels.
AFE + buffer	Chopper-stabilised instrumentation amplifier, programmable gain, driven-right-leg (DRL) bias, lead-off detection, anti-alias filter, input buffer.	Input-referred noise below 1 uVrms (0.5 to 100 Hz); input impedance above 100 Mohm; DC offset rejection.
ADC	Simultaneous-sampling delta-sigma converter, one modulator per channel (or a shared SAR with mux for low-channel variants).	24-bit, 250 Sps to 4 kSps per channel. Example: 32 ch x 1 kSps x 24 bit = 768 kbit/s.
DSP buffer	Rate-matching FIFO and ping-pong buffer between the converter clock domain and the DSP clock domain.	Async FIFO with CDC; AXI4-Stream master/slave; overflow and underrun flags.
Digital filter	Band-pass (0.5 to 100 Hz), 50 Hz mains notch (Indian grid), optional common-average re-reference, adaptive (LMS) artifact removal using EOG / IMU references.	FIR / IIR biquad cascade, fixed-point 24-bit data path, coefficients via AXI4-Lite.
SRAM buffer	Ring buffer holding recent history so that a flagged window can be delivered with pre-trigger context.	Example: 30 s x 32 ch x 256 Sps x 4 B = 983,040 B (about 960 KiB), ping-pong organised.
RTL threshold	Hard-wired feature extraction and comparison: line-length, band-power ratios (delta / theta / alpha / beta), spectral entropy, amplitude and variance gates, N-of-M channel voting, hysteresis and hold-off.	Per-channel programmable thresholds via AXI4-Lite; single-cycle compare; wake-up IRQ and enable lines.
Normal path	Window judged normal: bypass, store only a compressed summary (for example delta + Rice coding) or a periodic heartbeat sample.	Low-rate export over BLE / storage; accelerators stay off.
Abnormal path	Wake-up controller ungates the accelerators, starts clocks, and triggers a DMA of the flagged window plus pre-trigger history to both accelerators in parallel.	Wake-to-inference latency target in the low-millisecond range (to be measured).
Edge-AI accelerator	Compact quantised network (CNN / LSTM, EEGNet-class) for fast event type classification (seizure, spindle, artifact, etc.).	INT8 systolic MAC array; thousands to a few hundred thousand parameters; weights resident in local SRAM.
Swin Transformer accelerator	Hierarchical vision transformer with shifted-window self-attention for spatio-spectral analysis. Input: spectrograms, scalograms or scalp topographic maps; for the tumour pathway, MRI / CT slices.	Swin-T class: about 28 M parameters and about 4.5 G MACs per 224 x 224 input (about 28 MB at INT8, so weights sit in MRAM / LPDDR; a pruned or distilled variant is needed for fully on-die storage).
Output fusion	Merges both accelerator outputs, resolves disagreement by confidence, renders the diagnostic image (heat-map overlay) and the problem text (class label plus templated findings).	Image over display / USB / BLE; text over UART / BLE. Results carry a confidence value and timestamp.

Figure 1. Brain Condition Detector SoC: system block diagram and data flow



Interface colours indicate the AMBA protocol used on each link. The RISC-V management core, wake-up controller and RTL threshold engine remain powered; the dashed green path moves only the flagged window from the SRAM buffer to the accelerators.

Figure 2. AMBA AXI4 interconnect topology and AXI4-Stream sample fabric



Masters (left) and slaves (right) meet at an AXI4 crossbar. The AXI4-Stream fabric (bottom) carries samples from the ADC to the accelerators; the DMA converts between AXI4-Stream and AXI4-Full at the SRAM buffer. All configuration registers hang off an AXI4-Lite bridge.

3. SEMICONDUCTOR IP BLOCKS AND AXI4 INTEGRATION

The tables below list every IP block required for the SoC, grouped by function. The last column names the AXI4 role: **AXIS** = AXI4-Stream, **AXI-F** = AXI4-Full, **AXI-L** = AXI4-Lite, **M** = master, **S** = slave. Blocks marked *(AO)* sit in the always-on domain.

3.1 Analog and mixed-signal IP

IP block	Function and notes	AXI4 interface
Biopotential AFE array (AO)	Per-channel chopper instrumentation amp, programmable gain, DRL, lead-off and impedance measurement, electrode-offset cancellation.	Configured over AXI-L (via SPI-master wrapper); data out through the ADC block.
Optical AFE for fNIRS / PPG (AO)	LED drivers (two or more wavelengths), transimpedance amplifier, ambient-light cancellation, synchronous detection.	AXI-L config; samples to AXIS via aux ADC.
24-bit delta-sigma ADC bank (AO)	Simultaneous sampling per channel; digital decimation sinc / FIR stage; selectable output data rate.	AXIS master (samples); AXI-L slave (rate, gain, calibration).
Aux ADC / SAR (AO)	Slow auxiliary channels: temperature, EDA, respiration belt.	AXIS master; AXI-L slave.
PLL / clock generator and clock gating	Sample clock (low jitter), DSP clock, accelerator clocks with glitch-free gating.	AXI-L slave (dividers, gate enables).
Bandgap, LDOs, bias generation	Low-noise supplies for the AFE; separate rails for digital and gated domains.	AXI-L slave (trim, enable) via PMU.
Electrode / patient-isolation front-end	ESD, defibrillation protection, input protection, leakage-current limiting for IEC 60601 compliance.	Passive / off-chip; status via AXI-L GPIO.

3.2 Digital signal chain IP (always-on)

IP block	Function and notes	AXI4 interface
ADC deserializer / sample formatter (AO)	Converts SPI / LVDS ADC frames to channel-tagged samples with TID = channel and a sample-valid strobe.	AXIS master.
DSP buffer (async FIFO, ping-pong) (AO)	Clock-domain crossing and rate matching between converter and DSP clocks.	AXIS slave in, AXIS master out; AXI-L (watermarks, flags).
Digital filter bank (AO)	FIR / IIR band-pass, 50 Hz notch, common-average reference, LMS artifact canceller; coefficient RAM.	AXIS in / out; AXI-L slave for coefficients and mode.
Feature extractor (AO)	Streaming FFT / Goertzel band-power, line-length, variance, spectral entropy, per-window.	AXIS in, feature AXIS out to threshold; AXI-L config.
SRAM buffer controller (AO)	Ring buffer with write pointer, read pointer, pre-trigger lock; ECC optional.	AXI-F slave (to DMA); AXI-L slave (pointers, status).
AXI DMA (S2MM and MM2S) (AO)	Moves samples AXIS to SRAM (S2MM) and SRAM to AXIS / accelerators (MM2S); scatter-gather for window delivery.	AXI-F master; AXIS slave (S2MM) and master (MM2S); AXI-L slave (descriptors).
RTL threshold engine (AO)	Comparators, N-of-M vote, hysteresis, hold-off, wake request generation; per-channel limits.	AXIS slave; AXI-L slave (Sec. 4.4); IRQ and wake lines.
AXIS broadcaster / switch	Replicates the flagged window stream to both accelerators in parallel with backpressure handling.	AXIS slave in, two AXIS masters out; AXI-L (routing).

3.3 Accelerator IP (power-gated)

IP block	Function and notes	AXI4 interface
Edge-AI NPU	INT8 systolic MAC array, activation unit (ReLU / sigmoid / softmax LUT), pooling, local weight and activation SRAM, layer sequencer.	AXIS slave (input window); AXI-F master (weights, spill); AXI-L slave (CSR, layer config); IRQ.
Swin Transformer accelerator	Patch-embedding engine, window partition / shift (cyclic roll) address generator, QKV projection, windowed attention (Q x K-transpose, softmax, x V), relative position bias, MLP, LayerNorm, GELU, patch-merging, INT8 / FP16 MAC array.	AXIS slave (tokens / image); AXI-F master (weights, KV / attention tiles, feature maps); AXI-L slave (CSR); IRQ.
Spectrogram / scalogram generator	STFT or wavelet front-end converting EEG windows into time-frequency images for the Swin input.	AXIS in, AXIS out; AXI-L config.

IP block	Function and notes	AXI4 interface
Imaging ingest (tumour pathway)	DICOM / NIfTI slice receiver, normalisation and resize for MRI / CT slices from a host.	AXI-F master (writes slices to LPDDR); AXI-L slave.
Accelerator scratchpad SRAMs	Local tile memories for each accelerator; power-gated with the domain.	AXI-F slave (host / DMA preload); local ports to compute.
Output fusion / report generator	Confidence merge, saliency overlay renderer, template-based text generator (class label to finding string).	AXI-F master (frame buffer); AXIS (image out); AXI-L slave (CSR).

3.4 Processor, interconnect, memory and system IP

IP block	Function and notes	AXI4 interface
RISC-V management core (AO)	32-bit core (for example a VeeR EL2-class design) running boot, scheduling, power sequencing and result handling.	AXI-F master (instruction / data); AXI-L via bridge.
AXI4 crossbar (N x M)	Central interconnect with QoS, ID remap, outstanding-transaction tracking; connects all masters to all slaves.	AXI-F slave ports from masters, master ports to slaves.
AXI data-width and protocol converters	32 / 64 / 128 / 256-bit width conversion; AXI4 to AXI4-Lite bridge; AXI-to-APB bridge for slow peripherals.	AXI-F and AXI-L on both sides.
AXI async CDC bridges	Safe crossing between always-on, DSP and accelerator clock domains.	AXI-F / AXI-L both sides.
AXI firewall / protection unit	Per-master address and permission filtering; blocks rogue accesses to security and threshold registers.	AXI-F / AXI-L filter, AXI-L config.
Boot ROM and secure boot	Immutable first-stage loader, signature check of firmware and model weights.	AXI-F slave.
On-chip SRAM (ring buffer + system RAM)	Low-leakage SRAM macros with retention mode for the always-on domain.	AXI-F slave.
LPDDR controller and PHY	External memory for Swin weights, feature maps, frame buffers.	AXI-F slave (to controller); PHY native.
QSPI / eMMC NVM controller; eFlash / MRAM macro	Non-volatile storage of weights, logs, patient records.	AXI-F slave; AXI-L CSR.
Power management unit (PMU) (AO)	Power islands, DVFS, retention control, wake sequencing, brown-out detection.	AXI-L slave; power-switch controls; wake input from threshold.
Interrupt controller (PLIC) (AO)	Collects IRQs from threshold, DMA, accelerators.	AXI-L slave.
Timers, watchdog, RTC (AO)	Timestamping of events, watchdog for safety, low-power wake timer.	AXI-L slave.

3.5 Connectivity, security, safety and test IP

IP block	Function and notes	AXI4 interface
BLE / low-power radio MAC + baseband interface	Wireless export of summaries and reports.	AXI-F master (DMA), AXI-L slave.
USB 2.0 / Ethernet MAC	Wired offload, imaging ingest, firmware update.	AXI-F master, AXI-L slave.
Display controller (MIPI-DSI / parallel RGB)	Shows the diagnostic image and text report.	AXI-F master (frame read), AXI-L slave.
UART, SPI, I2C, GPIO	Text output, sensor configuration, LEDs, buttons.	AXI-L slave via bridge.
AES-256-GCM crypto engine, SHA-256, TRNG	Encrypts patient data at rest and in transit; supports secure boot.	AXI-F / AXI-L slave; AXIS for streaming encryption.
Hardware root of trust, key storage	Device identity, key wrapping, anti-rollback counter.	AXI-L slave behind firewall.
Safety and diagnostics monitors	ECC / parity checkers, watchdog, voltage and temperature monitors, lock-step option on the management core.	AXI-L slave; fault IRQs.

IP block	Function and notes	AXI4 interface
DFT: scan, MBIST, JTAG TAP, JTAG-to-AXI	Manufacturing test and bring-up debug; AXI master for register access.	AXI-F / AXI-L master (debug).
Trace / on-chip logic analyser	Debug visibility of AXIS and AXI transactions during FPGA prototyping.	AXIS / AXI monitor (passive).

4. AMBA AXI4 INTEGRATION DETAILS

4.1 How each AXI4 variant is used

Protocol	Used for	Typical configuration and rules
AXI4-Stream	Continuous sample and feature flow: ADC to FIFO to filter to feature extractor to DMA; window delivery from DMA to the RTL threshold, the broadcaster and both accelerators.	TDATA 24 / 32 b per sample (channel-interleaved), TVALID / TREADY back-pressure, TLAST at window end, TKEEP for partial beats, TID = channel number, TUSER = artifact / lead-off / saturation flags. No address channel: simple point-to-point or through an AXIS switch / broadcaster.
AXI4-Full (memory-mapped)	Burst traffic: DMA to SRAM ring buffer, accelerator weight and feature-map fetch from LPDDR / NVM, frame-buffer read by the display DMA, firmware fetch.	Five channels (AW, W, B, AR, R); INCR bursts up to 256 beats; 64 to 256-bit data; IDs for out-of-order response; QoS (AxQOS) giving the accelerators and DMA priority over debug; outstanding transactions for latency hiding; 4 KB boundary rule respected by DMA.
AXI4-Lite	Control and status plane: every block's CSR set (thresholds, filter coefficients, DMA descriptors, accelerator layer config, PMU, security).	32-bit data, single beat, no bursts, no IDs; reached through an AXI4-to-AXI4-Lite bridge off the crossbar; write strobes (WSTRB) honoured; interrupts routed through the PLIC.

4.2 Master and slave connectivity

Master	Slaves it must reach	Protocol and purpose
M0 RISC-V core	Boot ROM, SRAM, LPDDR, NVM, all CSRs	AXI-F for code / data; AXI-L for CSR programming.
M1 AXI DMA	SRAM, LPDDR	AXI-F bursts; AXIS on the other side. S2MM writes samples; MM2S reads the flagged window.
M2 Edge-AI NPU	SRAM (input), accelerator scratchpad, NVM (weights), result registers	AXI-F; result written back to a mailbox in SRAM.
M3 Swin accelerator	LPDDR (weights, feature maps), scratchpad, SRAM	AXI-F, high bandwidth (128 to 256 bit), QoS-prioritised.
M4 Output / display DMA	LPDDR / SRAM frame buffer, text mailbox	AXI-F read-only.
M5 JTAG-to-AXI	All slaves (debug window; firewall-gated in production)	AXI-F and AXI-L single beats.

4.3 Example address map (32-bit)

Base address	Size	Region	Interface
0x0000_0000	64 KiB	Boot ROM	AXI-F
0x1000_0000	1 MiB	On-chip SRAM (sample ring buffer + system RAM)	AXI-F
0x1100_0000	2 MiB	Accelerator scratchpads (NPU + Swin)	AXI-F
0x4000_0000	64 KiB	AFE / ADC control (SPI-master wrapper)	AXI-L
0x4001_0000	64 KiB	Digital filter and feature extractor CSR	AXI-L
0x4002_0000	64 KiB	RTL threshold engine CSR (Sec. 4.4)	AXI-L
0x4003_0000	64 KiB	Wake-up controller / PMU	AXI-L
0x4004_0000	64 KiB	AXI DMA CSR and descriptors	AXI-L
0x4005_0000	64 KiB	Edge-AI NPU CSR	AXI-L
0x4006_0000	64 KiB	Swin accelerator CSR	AXI-L
0x4007_0000	64 KiB	Output fusion and display CSR	AXI-L
0x4008_0000	64 KiB	Security block (AES, SHA, TRNG, root of trust)	AXI-L
0x4009_0000	64 KiB	UART / SPI / I2C / GPIO / timers / PLIC	AXI-L
0x6000_0000	128 MiB	QSPI NVM (execute-in-place window)	AXI-F
0x8000_0000	up to 1 GiB	LPDDR (Swin weights, feature maps, frame buffers)	AXI-F

4.4 RTL threshold engine register map (AXI4-Lite, base 0x4002_0000)

Offset	Register	Description
0x00	CTRL	Enable, soft reset, mode (monitor / calibrate), window-length select.
0x04	STATUS	Abnormal flag, window counter, FIFO overflow, last-trigger channel.
0x08	CH_MASK	Per-channel enable mask for the voting logic.
0x0C	LL_THR	Line-length threshold (per window).
0x10	BP_RATIO_THR	Band-power ratio limits (for example theta / alpha, delta / total).
0x14	ENT_THR	Spectral-entropy limit.
0x18	VOTE_NM	N-of-M channel voting configuration.
0x1C	HYST_HOLD	Hysteresis and hold-off counts before returning to normal.
0x20	PRETRIG_LEN	Seconds of pre-trigger history to deliver with a flagged window.
0x24	IRQ_EN	Interrupt enable bits.
0x28	IRQ_STATUS	Interrupt status (write-1-to-clear).
0x2C	WAKE_TARGET	Bit 0: wake Edge-AI NPU; bit 1: wake Swin accelerator; bit 2: wake both.

4.5 Integration and verification notes

- **Always-on boundary.** All AXI links crossing from the always-on domain into the gated domain need isolation cells and a clean reset / handshake: the AXI interface of a gated block must be quiesced (no outstanding transactions) before power-down, so the PMU waits for the accelerator's idle flag.
- **Clock-domain crossings.** The converter clock, DSP clock and accelerator clocks are separate; use async FIFOs on AXIS and AXI async bridges on memory-mapped paths.
- **Back-pressure.** The broadcaster must not drop samples when one accelerator stalls; use per-port skid buffers and a window-level credit scheme. The ADC side tolerates no stall, hence the ring buffer.
- **Latency and QoS.** Give the DMA and Swin master higher AxQOS than debug; reserve bandwidth for the display DMA to avoid frame tearing.
- **Verification.** Use AXI4 / AXI4-Lite / AXI4-Stream protocol-checker VIP in a UVM environment, constrained-random window generators for the threshold engine (including corner cases at threshold boundaries), and FPGA prototyping (for example on Vivado) to validate throughput before tape-out.

5. BIOMEDICAL SENSORS FOR THE FULL RANGE OF BRAIN AND NEUROLOGICAL DATA

No single chip or electrode type captures every neurological signal, so the sensors are grouped into three tiers. **Tier 1** connects directly to the on-SoC analog front end. **Tier 2** are wearable auxiliary sensors on slow serial interfaces. **Tier 3** are clinical or imaging modalities that cannot be worn and reach the SoC through a host interface for offline or triage analysis.

5.1 Tier 1: on-SoC biopotential front end

Sensor	What it measures	Bandwidth / rate	Conditions supported
Scalp EEG (10-20 / 10-10; wet Ag/AgCl, semi-dry or dry; 19 to 64 ch)	Cortical field potentials (delta to gamma).	0.1 to 100 Hz (to 500 Hz for HFO); 250 Sps to 1 kSps	Epilepsy, sleep staging, encephalopathy, dementia slowing, TBI / concussion, stroke asymmetry, migraine, ADHD, depression markers, disorders of consciousness.
EOG (2 to 4 electrodes)	Eye movement and blink potentials.	0.1 to 30 Hz	REM detection, eye-artifact reference, oculomotor abnormalities.
Surface EMG (chin, limbs)	Muscle activity and tone.	10 to 500 Hz	REM sleep behaviour disorder, periodic limb movements, tremor, myoclonus.
ECG	Heart electrical activity.	0.5 to 150 Hz	Autonomic assessment, seizure-related arrhythmia, cardiac artifact removal.
Evoked-potential acquisition (ABR / VEP / SSEP)	Time-locked EEG responses to auditory, visual or somatosensory stimuli (uses the EEG channels plus a stimulus trigger input).	0.1 Hz to 3 kHz (ABR) with averaging	Multiple sclerosis, optic and auditory pathway disorders, coma prognosis.

5.2 Tier 2: wearable auxiliary sensors

Sensor	What it measures	Bandwidth / rate	Conditions supported
3-axis accelerometer + gyroscope (IMU)	Head and limb motion, tremor, gait, falls.	50 to 200 Hz	Convulsive seizures, Parkinson's tremor, concussion impact, motion-artifact reference.
PPG / pulse oximeter (SpO2)	Pulse waveform, oxygen saturation.	50 to 200 Hz	Sleep apnea, seizure-related desaturation, autonomic changes.
Electrodermal activity (EDA / GSR)	Skin conductance (sympathetic arousal).	4 to 32 Hz	Convulsive seizure detection, stress and autonomic dysfunction.
Skin temperature sensor	Peripheral temperature.	1 Hz	Autonomic markers, fever context.
Respiration (thoracic belt / nasal airflow)	Breathing effort and flow.	10 to 50 Hz	Sleep-disordered breathing, central apnea.
fNIRS (about 760 / 850 nm, multi-distance)	Cortical haemoglobin oxygenation changes.	10 to 50 Hz	Stroke monitoring, cognitive load, subdural / intracranial haematoma screening, epilepsy haemodynamics.
Microphone / contact audio	Speech, snoring, vocalisation.	up to 16 kHz	Speech biomarkers (dysarthria, Parkinson's), snoring, seizure vocalisation.
Eye tracker / pupillometer (IR camera)	Gaze, saccades, pupil reflex.	60 to 120 fps	Concussion screening, Parkinsonian oculomotor signs, raised intracranial pressure signs.

5.3 Tier 3: clinical and imaging modalities (external or invasive)

Modality	What it measures	Interface to the SoC	Conditions supported
Intracranial EEG (ECoG / sEEG)	Direct cortical / depth recordings; high-gamma to 500+ Hz.	Separate implantable front end; clinical use only.	Epilepsy surgery planning, seizure focus mapping.
MEG (including OPM-MEG)	Magnetic fields from neural currents (fT scale).	Offline data ingest (USB / Ethernet).	Source localisation, epilepsy, functional mapping.
Transcranial Doppler ultrasound (TCD)	Cerebral artery blood-flow velocity.	External module, serial link.	Stroke, vasospasm, stenosis, emboli.

Modality	What it measures	Interface to the SoC	Conditions supported
Electrical impedance tomography (EIT)	Impedance map of the head (research stage).	Dedicated AFE or external module.	Stroke type differentiation, oedema monitoring.
Intracranial pressure (ICP) probe	Pressure inside the skull.	Invasive sensor, aux ADC / serial.	TBI, hydrocephalus, critical care monitoring.
MRI / CT / PET (DICOM, NIfTI)	Structural and metabolic brain imaging.	Host interface into the imaging-ingest IP, then to the Swin accelerator.	Brain tumours, haemorrhage, infarct, atrophy. This is the primary route for tumour screening.
Lab biomarkers (for example NfL, GFAP, p-tau)	Blood / CSF markers.	Entered as patient metadata by the host.	Context for neurodegeneration and injury.

5.4 Condition-to-modality summary

Condition	Primary modality	Supporting modalities	SoC compute path
Epilepsy / seizures	EEG	IMU, EDA, ECG, PPG, EMG	Threshold, then Edge-AI (event type), Swin (focus map)
Sleep disorders	EEG + EOG + EMG	ECG, respiration, SpO2, PPG, microphone	Threshold on sleep-stage anomalies, then Edge-AI + Swin
Alzheimer's / dementia	EEG (slowing, coherence)	Evoked potentials, speech audio, eye tracking	Threshold on slowing, then Swin (spectrogram)
Parkinson's disease	IMU + EMG	EEG, speech audio, eye tracker	Edge-AI (tremor), Swin (trends)
Stroke	EEG + fNIRS	TCD, EIT, MRI / CT (clinical)	Threshold on asymmetry, then Swin
TBI / concussion	EEG + IMU	Eye tracker, ICP, fNIRS	Threshold, then Edge-AI + Swin
Brain tumour	MRI / CT (external)	EEG focal slowing (suggestive only), EIT	Swin on imaging slices (triage support)
Migraine / ADHD / depression	EEG	ECG, EDA, eye tracker	Swin (spectral biomarkers); screening only
Disorders of consciousness / encephalopathy	EEG + evoked potentials	ECG, SpO2, EMG	Threshold, then Edge-AI + Swin

6. SUGGESTED IMPLEMENTATION ROADMAP

- **Phase 1, RTL + FPGA:** implement the always-on chain (filter, SRAM buffer, threshold engine) with AXI4-Stream and AXI4-Lite, and validate with recorded public EEG datasets (for example CHB-MIT, Sleep-EDF) replayed through the stream interface.
- **Phase 2, accelerators:** train and quantise the Edge-AI model first, then a reduced Swin variant; prototype the NPU and Swin engine on FPGA behind the AXI4 interconnect; measure latency and bandwidth.
- **Phase 3, system integration:** add the RISC-V core, PMU and power-domain gating; verify wake-up sequencing and AXI quiescing under UVM.
- **Phase 4, implementation:** synthesis, power analysis (always-on versus gated), analog AFE co-design, DFT insertion, and regulatory documentation planning.